

WISH - 2026 - Fourier and Laplace Transforms

* Full Name

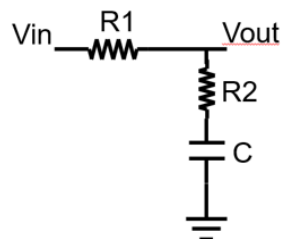
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WiSH26007

Q1. What is the maximum phase difference between input and output across all input frequencies given $R1=10R2$

- a) 0°
- b) Between 0° to 45°
- c) Between 45° to 90°
- d) 90°



Answer the Question 1

☐ a

☐ b

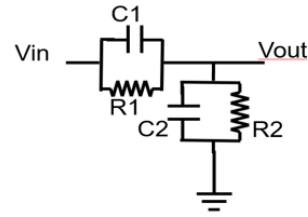
☒ **c**

☐ d

Q2. 4 statements are given for the circuit below

1. $|H(0)| > |H(\infty)|$ for $R1=R2$, $C1 > C2$
2. $|H(0)| > |H(\infty)|$ for $R1=R2$, $C1 < C2$
3. $|H(0)| > |H(\infty)|$ for $C1=C2$, $R1 > R2$
4. $|H(0)| > |H(\infty)|$ for $C1=C2$, $R2 > R1$

- a) 1 and 3 correct
- b) 2 and 4 correct
- c) 1 and 4 correct
- d) Only 3 correct



Answer the Question 2

☐ a

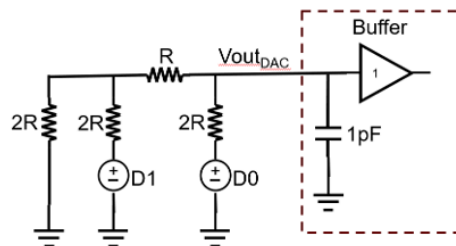
☐ c

☒ b

☐ d

Q3. The output of a 2-bit R-2R DAC goes to a buffer that has 1pF input capacitor. The input of the DAC changes every 1ns, what max value of R can be used if 99% settling accuracy is required

- a) 123 Ohm
- b) 217 Ohm
- c) 345 Ohm
- d) 491 Ohm



Answer the Question 3

☐ a

☐ c

☒ b

☐ d

Q4. A1 and A2 are ideal buffers with a gain of 2 and 5, respectively.

$R1 = 10\Omega$

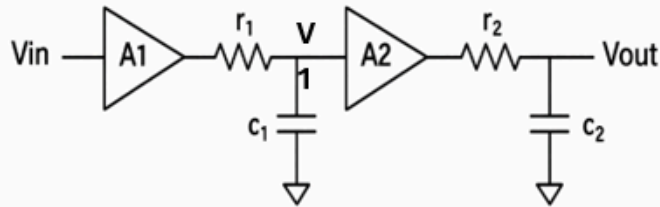
$R2 = 100\Omega$

$C1 = 0.1F$

$C2 = 1F$

Determine the following

- 3dB point at **V1** and **Vout** (Closest approximate)
- Magnitude and phase of **Vout** at $\omega = 100$ rad/s



- 0.01 rad/s and 1 rad/s (ii) -100 dB and -180°
- 0.01 rad/s and 0.01 rad/s (ii) -100 dB and -180°
- 0.01 rad/s and 0.01 rad/s (ii) -100 dB and 180°
- 0.01 rad/s and 0.01 rad/s (ii) -80 dB and -180°
- 1 rad/s and 1 rad/s (ii) -80 dB and -90°
- 0.01 rad/s and 0.01 rad/s (ii) -80 dB and -90°

Answer the Question 4

☐ a

☐ d

☒ b

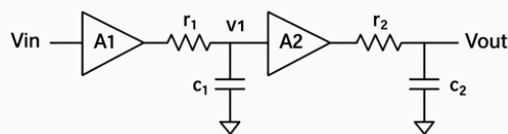
☐ e

☐ c

☐ f

Q5. In the above question (4), assume **A1 and A2 are non ideal buffers**. They have an input resistance of infinity but an **output resistance of 10Ω** . It is required by design that even after accounting for this non-ideality, the magnitude of **Vout** at $\omega = 100$ rad/s remain the same as in Q1. By how much should R2 be changed to get the same rejection as Q1.

- Increase R2 by 10 ohms
- Increase R2 by 60 Ohms
- Increase R2 by 100 ohms
- Increase R2 by 40 ohms
- Decrease R2 by 10 ohms
- Decrease R2 by 60 ohms
- Decrease R2 by 100 ohms
- Decrease R2 by 40 ohms
- Do not change R2



Answer the Question 5

☐ a

☒ f

☐ b

☐ g

☐ c

☐ h

☐ d

☐ i

☐ e

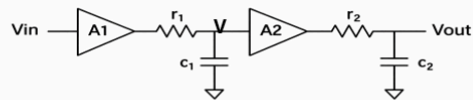
Q6. A1 and A2 are ideal buffers with a gain of 2 and 5, respectively.

R1 = 10Ω

R2 = 1Ω

C1 = 0.1F

C2 = 1F



Determine the following

3dB point at **V1** and **Vout** (Closest approximate)

Phase of **Vout** when $\omega = \omega_{p1}$ (Stage 1 pole)

- a). (i) 1 rad/s and 1 rad/s (ii) -45°
- b). (i) 1 rad/s and 1 rad/s (ii) -90°
- c). (i) 1 rad/s and $\sqrt{1.414}$ rad/s (ii) -45°
- d). (i) 1 rad/s and $\sqrt{1.414}$ rad/s (ii) -90°
- e). (i) $\sqrt{0.414}$ rad/s and $\sqrt{0.414}$ rad/s (ii) -90°
- f). (i) $\sqrt{0.414}$ rad/s and $\sqrt{0.414}$ rad/s (ii) -45°
- g). (i) $\sqrt{1.414}$ rad/s and $\sqrt{1.414}$ rad/s (ii) -90°
- h). (i) $\sqrt{1.414}$ rad/s and $\sqrt{1.414}$ rad/s (ii) -45°
- i). (i) 1 rad/s and $\sqrt{0.414}$ rad/s (ii) -90°
- j). (i) 1 rad/s and $\sqrt{0.414}$ rad/s (ii) -45°

Answer the Question 6

☐ a

☐ f

☐ b

☐ g

☐ c

☐ h

☐ d

☒ i

☐ e

☐ j

Q7. Consider the RLC circuit shown below:

$$R = 1000\Omega$$

$$C = 1/900F$$

$$L = 1H$$

A periodic square pulse is given as input to the circuit. It has a duty cycle of 50% and has a time period of $\frac{\pi}{5} (T_0)$. What would the output waveform (Vout) largely look like ?

- A. A distorted square pulse with a high rise or fall time



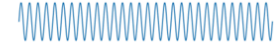
- B. An un-distorted square pulse (Output waveform follows the input)



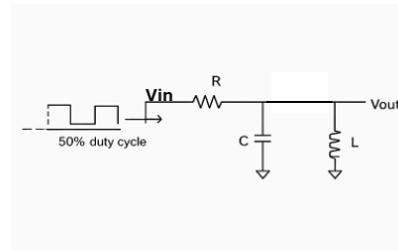
- C. A sine wave corresponding to a frequency of ω_0



- D. A sine wave corresponding to a frequency of $3\omega_0$



- E. A series of impulses centered about origin



Answer the Question 7

☐ a

☒ d

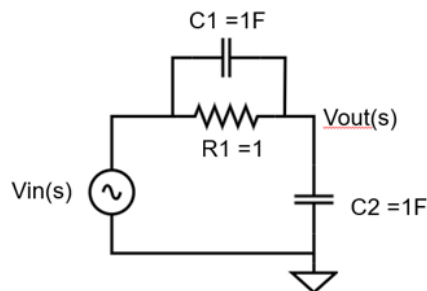
☐ b

☐ e

☐ c

Q8. In the below fig, what is the gain of Vout/Vin (s) at 10 rad/s?

- a) 0dB
b) -20dB
c) -6dB
d) -10dB



Answer the Question 8

☐ a

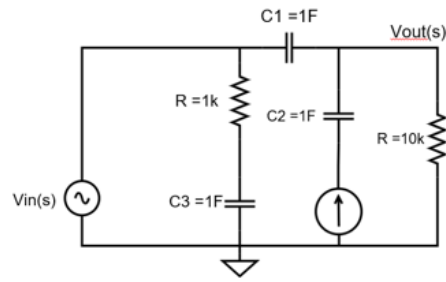
☒ c

☐ b

☐ d

Q9. In the below fig, find the number of poles and zeroes in the transfer function $V_{out}(s)/V_{in}(s)$?

- a) 3poles,3zeroes
- b) 2 poles, 2zeroes
- c) 2 pole, 1 zero
- d) 1 pole, 1zero



Answer the Question 9

☐ a

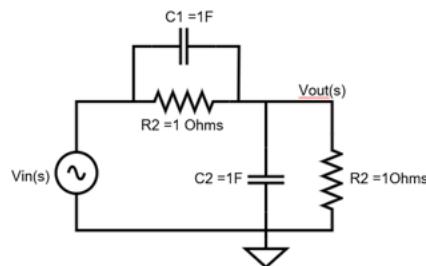
☐ c

☐ b

☒ **d**

Q10. In the below fig, what is the gain of V_{out}/V_{in} (s) at 0rad/s, 1 rad/s, 100 rad/s?

- a) 0.5,0.5,0.5
- b) 0.5,1,1
- c) 0.5,1,0.5
- d) 1,1,0.5



Answer the Question 10

☒ **a**

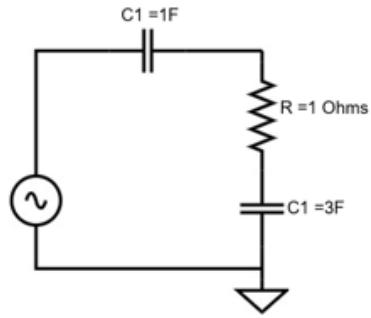
☐ c

☐ b

☐ d

Q11. In the below fig, what is the gain of V_{out}/V_{in} (s) at 100 rad/s?

- a) 0.5
- b) 1
- c) 0.33
- d) 0.25



Answer the Question 11

☐ a

☐ c

☒ **b**

☐ d

Done

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